

# Effects of co-exposure to 900 MHz radiofrequency electromagnetic fields and high-level noise on sleep, weight, and food intake parameters in juvenile rats

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## Abstract

Objective: Electrohypersensitive people attribute various symptoms to exposure of radiofrequency electromagnetic fields (RF-EMF); sleep disturbance is the most frequently cited. However, laboratory experiments have yielded conflicting results regarding sleep alterations. Our hypothesis was that exposure to RF-EMF alone would lead to slight or non-significant effects but that co-exposure to RF-EMFs and other environmental constraints (such as noise) would lead to significant effects.

Methods: 3-week-old male Wistar rats (4 groups, n=12 per group) were exposed for 5 weeks to continuous RF-EMF (900 MHz, 1.8 V/m, SAR=30 mW/kg) in the presence or absence of high-level noise (87.5 dB, 50-20000 Hz) during the rest period. After 5 weeks of exposure, sleep (24 h recording), food and water intakes, and body weight were recorded with or without RF-EMF and/or noise. At the end of this recording period, sleep was scored during the 1h resttime in the absence of noise and of RF-EMF exposure.

Results: Exposure to RF-EMF and/or noise was associated with body weight gain, with hyperphagia in the noise-only and RF-EMF+noise groups and hypophagia in the RF-EMF-only group. Sleep parameters recording over 24 h highlighted a higher frequency of active wakefulness in the RF-EMF-only group and a lower non-rapid eye movement/rapid eye movement sleep ratio during the active period in the noise-only group. There were no differences in sleep duration in either group. During the one-hour, constraint-free sleep recording, sleep rebound was observed in the noise-only group but not in the RF-EMF-only and RF-EMF+noise groups.

Conclusion: Our study showed effects of RF-EMF, regardless of whether or not the animals were also exposed to noise. However, the RF-EMF+noise group presented no exacerbation of those effects. Our results did not support the hypothesis whereby the effects of RF-EMF on physiological functions studied are only visible in animals exposed to both noise and RF-EMF.

Keywords: RF-EMF, noise, juvenile rat, sleep, food intake

## 1. Introduction

Various symptoms (headaches, fatigue, difficulty concentrating, skin manifestations, and tinnitus) have been reported with regard to (and then attributed to) radiofrequency electromagnetic field (RF-EMF) exposure in people living close to mobile phone base stations (Santini et al., 2002) and in people suffering from electrohypersensitivity (EHS) (Johansson et al., 2010; Knave et al., 1992). EHS is a phenomenon characterized by the perception from subjects of non-specific symptoms, without biological explanations and attributed by the subjects themselves at electromagnetic fields exposure. Although sleep disorders are the most frequently mentioned symptoms (Berg-Beckhoff et al., 2009), a possible effect of RF-EMF on sleep is subject to debate (Mohler et al., 2012). There is also controversy with regard to the results of intermittent exposure to RF-EMF (i.e. 15 on-off cycles at 900 MHz over 8 h) on adult in laboratory, which was associated with a significant decrease in rapid eye movement (REM) sleep but no changes in the other sleep states. In rodents, chronic RF-EMF exposure (5 weeks at 900 MHz 1.0 V/m) induced REM fragmentation (Pelletier et al., 2013), whereas acute exposure (at 1.8 or 2.45 GHz for 24 h) did not alter sleep (Crouzier et al., 2007).

Some studies involved co-exposure to RF-EMF and another environmental physical constraint (Pelletier et al., 2013) or chemical constraint (Occelli et al., 2018). The results showed that the effects of RF-EMF were amplified (to a variable extent) by the other factors, relative to RF-EMF exposure alone. In view of these observations and the ongoing debate, we checked that exposure to RF-EMFs alone, at the level typically encountered in everyday life (measurements in France showed a mean level around 0.36 V/m and 90% of measurements not exceeded 1.6 V/m, (ANFR 2018)), leads to small or non-significant effects, whereas synergy between RF-EMF and another environmental constraint would led to statistically significant physiological effects.

In our study, the other environmental constraint chosen was the noise. In fact, noise is a major everyday environmental constraint. People may be exposed to noise (and RF-EMFs) in the workplace and/or during leisure activities. It has been reported that noise disturbs sleep (Blois et al., 1980; Fruhstorfer et al., 1988) and metabolism (Axelsson and Lindgren, 1985; Pyko et al., 2015). Indeed, sleep perturbation may cause metabolic disorders (Spiegel et al., 1999).

Although exposure to noise or RF-EMF in everyday life is continuous (i.e. chronic), most laboratory studies investigated acute exposure. Moreover, most of these studies were performed on adults. In fact, juvenile life is a vulnerable period because of (i) the ongoing development of physiological and neurologic functions, and (ii) the greater RF-EMF

82 penetration into a juvenile' tissues (including the brain), relative to the adult (Wiart et al.,  
83 2011). To the best of our knowledge, the present study is the first to analyze the impact of co-  
84 exposure to RF-EMF and high-level noise. Other novel aspects of the present study are the  
85 chronic nature of the exposure, the fact that the experiments were performed on juvenile  
86 animals, and the recording different periods (including a non-exposed period after period of  
87 exposure).

## 2. Material and Methods

### 2.1 Animals

Experiments were conducted on 48 male Wistar rats (Janvier Labs, Le Genest Saint Isle, France) weighing between 55 and 85 g and aged 3 weeks at the time of their arrival in our facility (on day (D)1). Four groups of 12 animals were formed: a group exposed to noise only, a group exposed to an RF-EMF only, a group co-exposed to RF-EMF and noise, and a control (non-exposed) group. The groups were housed in anechoic chambers with a 12 h:12 h dark/light cycle (lights on at 6 am and off at 6 pm) and controlled thermoneutral air temperature ( $24 \pm 1$  °C), relative air humidity (mean  $\pm$  standard deviation (SD):  $39 \pm 12\%$ ), and air velocity ( $<0.2$  m/s). Rats were individually housed in plastic cages (425 mm x 266 mm x 185 mm) within the chamber. Standard chow (3436EXF12, Serlab, Montataire, France) and water were available *ad libitum*. Daily animal care was performed between 5 and 6 pm; the RF-EMF and noise were turned off during this period. All experiments were performed in accordance with the European guidelines (2010/63/EU) and the French governmental decree 2013-118 on the care and use of laboratory animals. The study protocol was approved by the nationally accredited Regional Directorate for Health, Animal and Environment Protection (Amiens, France) and the French Ministry of Research (license number: APAFIS#3735-2016012017118094 v3).

Depending on the group, exposure began on D5 and ran for 5 weeks until sacrifice (i.e. exposure from D5 to D41).

### 2.2 Exposure to an RF-EMF

The RF-EMF exposure started after four days of acclimatization. The climatic chambers were equipped with RF-EMF antennas (model 800–10465, KATHREIN-Werke KG, Rosenheim, Germany) powered by a generator (model RFS 900–64, RFPA, Artigues-près-Bordeaux, France) emitting a 900 MHz continuous-wave EMF. Antennas were located horizontally in the climatic chamber, 80 cm above the exposed rats' boxes. The generator's power was set to obtain a field intensity of  $1.8 \pm 0.6$  V/m. The animals' specific absorption rate was calculated with a mean value of 30 mW/kg (calculated at 3 and 8 weeks of age for three parts of the animal: head, body, and tail). The level of RF-EMF exposure was checked once a week with a radiofrequency probe (PMM EP600, Narda Safety Test Solution, Hauppauge, NY, USA) and monitored with computer software (Win EP 600, Narda Safety Test Solution). The RF-EMF exposure ran for 23 h per day.

### 2.3 Noise exposure

Noise-exposed animals were subjected to a 24 h soundtrack divided into two main periods: a noisy light period (the rat's rest period) from 6 am to 5 pm, and a noise-free dark period from 6 pm to 6 am. During the noise-free period and the handling period, the intensity of the background noise was 65 dB.

The noise exposure period was split into 10 min segments, each of which had a unique combination of noise types, frequencies and intensities. The noise types were urban sounds (traffic, roadworks, sirens, etc.), music, and artificial sounds (white noise, red noise, sinusoids, etc.). The mean  $\pm$  SD (range) sound level was  $87.5 \pm 3.7$  dB (59–111). The frequencies ranged from 50 to 20,000 Hz. To avoid habituation, four different 24 h noise exposure files were built out of 10-minute segments arranged in a pseudo-random order. Each day's noise exposure file was chosen in a pseudo-random manner. The noise was regulated by an amplifier (Combo 130, Audiophony Hit Music SAS, Fontanes, France) and delivered through loudspeakers (K50-8 Ohm, JBSystems, Groot-Bijgaarden, Belgium).

### 2.4 Protocol

After 4 days of habituation to the housing conditions, animals were exposed (or not) to noise and/or RF-EMF for 5 weeks (until D41). On D26 or D27, a telemetric sensor was surgically implanted in each animal. From D33 to D39, food intake, water intake and body weight data were recorded. Electroencephalography (EEG) and electromyography (EMG) were recorded for one day (D34) with exposure and for one hour (D39) without exposure. At D41, the rats were sacrificed by a heart puncture under general anesthesia with a mixture of air and 2.5% isoflurane (Iso-Vet 1000 mg/g, Piramal Healthcare UK Ltd, Morpeth, United Kingdom).

### 2.5 Surgery

At the beginning of the 4<sup>th</sup> week, a subcutaneous telemetric EEG and EMG sensor (F20-EET, Data Sciences International, St. Paul, MN, USA) was implanted under general anesthesia with isoflurane (5% for induction, and then 2.5% during surgery). To record the EEG signal, two gold-plated screws (15 SURTEX, Dentatus AB, Spånga, Sweden) were inserted into the animal's skull, just above the dura matter. Dental cement (Dentalon, Henri Schein, Alfortville, France) was used to fix the probe and isolate the EEG signal. The EMG signal was recorded from two probes were inserted into the dorsal muscles of the neck. After surgery, animals were allowed to recover for 5 days with or without exposure in accordance with their group.

## 2.6 Data acquisition and analysis

### 2.6.1 Body weight

The animals' body mass was measured every day using scales (Scaltec SPO-62, Scaltec Instruments, Göttingen, Germany; sensitivity: 0.1 g). Due to difference of weight between the 4 groups of animals, the weight gain (in grams) was calculated as follows:  $\text{weight gain} = \text{Weight at } D_x - \text{Weight at } D_1$ .

### 2.6.2 Food and water intakes

Daily food intake was scored automatically with individual scales (Sartorius TE601, Sartorius Lab Instruments GmbH; sensitivity: 0.1 g). The time, the duration (in seconds) and the quantity of each meal (in grams) were recorded using in-house software. Water intake was measured daily, using individual nursing bottles (sensitivity: 5 mL). The ratios between the total quantity of food eaten and the animal's body weight was calculated.

Due to technical problems, food and water intakes was not calculated for 1 of the 12 animals in the RF-EMF group. The food and water intakes were calculated for the other 11 animals in the RF-EMF group, and for all animals in the noise-exposed, RF-EMF+noise, and control groups.

### 2.6.3 Sleep

The EEG and EMG sensors were connected wirelessly to a receiver (RPC-1, Data Sciences International) and a computer via a matrix (Data Exchange Matrix, Data Sciences International). Data were recorded using Ponemah software (Data Sciences International).

Recordings were scored visually every 4 seconds with Spike2 software (version 7.01, Cambridge Electronic Design, Cambridge, UK) as active wakefulness (AW), quiet wakefulness (QW), non-rapid-eye-movement (NREM) sleep, and REM sleep. Quiet wakefulness was defined as a short episode (lasting between 8 s and 2 min) of wakefulness within a sleep episode, and during which the animal did not move or eat (Pelletier et al., 2013). For NREM and REM sleep, the total amount (in min per 24 h), episode frequency (per hour), mean episode duration (in min) and proportion of total sleep time (TST, in %) were calculated.

Due to occasional technical problems, sleep parameters were calculated for 11 of the 12 animals in the RF-EMF-only group, 10 of the 12 animals in the noise-only group, all 12 animals in the RF-EMF+noise group, and all 12 animals in the controls.

### 2.6.4 Statistical analysis

185 Data (expressed as the mean  $\pm$  SD) were analyzed using Statview software (version 5.0, SAS  
186 Institute Inc., Cary, NC, USA). When the data were normally distributed (according to a  
187 Kolmogorov-Smirnov test), a two-way analysis of variance (with Fisher's partial least-squares  
188 difference post-test) was applied. Simple effects of RF-EMF exposure (i.e. RF-EMF  $\pm$  noise  
189 vs no RF-EMF  $\pm$  noise), of noise exposure (i.e. noise  $\pm$  RF-EMF vs no noise  $\pm$  RF-EMF) and  
190 interaction RF-EMF x noise were analyzed. If the data were non normal (e.g. food and water  
191 intake parameters), a Kruskal-Wallis and a Mann-Whitney tests were used. Comparisons of  
192 the RF-EMF+noise group vs. the control group were also performed using a non-parametric  
193 test. The threshold for statistical significance was set to  $p < 0.05$ .



### 3. Results

#### 3.1 The animals' body weight

The body weight gain (related to weight on the day of arrival) of the RF-EMF-exposed, noise-exposed and RF-EMF+noise animals was higher than that in controls (control:  $210.5 \pm 25.5$  g, RF-EMF-exposed:  $231.9.5 \pm 27.6$  g,  $p < 0.001$  vs controls; noise-exposed:  $222.7 \pm 28.8$  g,  $p = 0.018$  vs controls; and RF-EMF+noise:  $223.5 \pm 23.3$  g,  $p = 0.005$  vs controls). Body weight gain was significantly lower in the noise-exposed group than in the RF-EMF exposed group ( $p = 0.037$ ).

#### 3.2 Food and water intake on 24 h

**Animals exposed to noise only** ate more than controls did, regardless of the nycthemeral period. This result was due to a higher total quantity of food, a greater number of meals, and a longer mean meal duration. Moreover, the noise-exposed animals drank more than controls did (Table 1).

In contrast, **animals exposed to an RF-EMF only** ate less than controls due to a lower number of meals and a shorter mean meal duration. However, the quantity of food relative to the body weight is similar to controls. As in the noise-only group, the animals in the RF-EMF-only group drank more than controls did.

**The animals in the RF-EMF+noise** group tended to eat more than controls did, due to a higher mean meal quantity, a longer mean meal duration but a lower number of meals over 24 h. Compared with the noise-only group, animals in the RF-EMF+noise group ate less food in total and had fewer meals. Compared with animals exposed to RF-EMF only, the total quantity and meal duration were higher in animals in the RF-EMF+noise group. The water intake for RF-EMF+noise animals did not differ significantly from that of the controls, but was lower than in animals exposed to noise only or an RF-EMF only.

#### 3.3 Sleep measured over 24 h with exposure

The **effects of noise** (independently of RF-EMF exposure, significant interaction between noise exposure and light/dark periods) were only significant during the dark period (i.e. during the recovery from noise exposure) (Table 2): the total durations of NREM sleep and QW were lower in the noise group ( $-22.6$  min,  $p = 0.01$  and  $-6.8$  min,  $p = 0.03$ , respectively) and tended to be compensated by a higher total duration of AW and REM ( $+24.2$  min,  $p = 0.08$  and  $+5.2$  min,  $p = 0.1$ , respectively), so that the relative duration of NREM sleep (relative to the TST) fell and the relative duration of REM sleep rose ( $p = 0.004$ ).

**Exposure to RF-EMF** (independently of noise exposure, significant effect of RF-EMF exposure alone) significantly induced a higher frequency ( $p=0.029$ ) and a shorter mean duration ( $p=0.012$ ) of AW episodes, so that the total duration did not differ significantly. In contrast to the effects of noise, the results in the RF-EMF group were not dark/light-dependent.

**The parameters in the RF-EMF+noise group** did not significantly differ from those measured in the three other groups. Interestingly, the only sleep parameters influenced by a significant interaction between the noise exposure and the RF-EMF exposure was the mean duration of NREM sleep episodes: in contrast to the RF-EMF+noise group, the RF-EMF exposure alone and the noise exposure alone had longer NREM sleep episodes (respectively  $p=0.048$  and  $p=0.006$  vs the control group) (Table 2).

### *3.4 Sleep measured over 1 h without exposure*

**Noise exposure** (independently of RF-EMF exposure, significant effect of noise exposure alone) induced significant effects during the 1 h-long, noise- and RF-EMF-free in the daytime (Table 3). In the noise-only group, the TST was significantly longer ( $p=0.036$ ) as a consequence of a longer duration of NREM ( $p=0.05$ ) and REM ( $p=0.05$ ) and a shorter AW duration ( $p=0.02$ ), despite a slight increase in the total duration of QW ( $p=0.014$ ). There were no statistically significant differences in REM and NREM but there was a trend towards a higher percentage of REM sleep and a lower percentage of NREM sleep, relative to animals not exposed to noise.

**Exposure to an RF-EMF alone** did not lead to differences vs. the control group.

**Sleep variables in the RF-EMF+noise group** did not differ from those measured in the noise-only or RF-EMF-only groups.

## 4. Discussion

### 4.1 Weight

Our results showed that whatever the type of exposure (RF-EMF and/or noise), exposed animals gained more body weight than controls. The RF-EMF-only group gained significantly more weight than the noise-only group. The association between higher body weight and noise exposure has already been described in rodent models (Mavanji et al., 2013; Parrish and Teske, 2017). Likewise, RF-EMF exposure induced weight variations (+6%) in an adult rodent model after chronic exposure (280 days) to a 900 MHz RF-EMF (intensity not indicated) (Sommer et al., 2004); however, the RF-EMF literature is very sparse in this respect. Despite a greater body weight gain in animals in the RF-EMF+noise group, we did not observe a specific effect of co-exposure to an RF-EMF and noise; since there were no significant differences between the RF-EMF+noise group on one hand and the RF-EMF-only group and noise-only groups on the other. Several literature studies have demonstrated that stress and/or loss of sleep can lead to weight gain and changes in food intake (von Kries et al., 2002; Michel et al., 2003; Spiegel et al., 2004). Even though the sleep alterations associated with our environmental exposures were small (see below), we cannot rule out a possible involvement in the greater body weight in the RF-EMF+noise group.

### 4.2 Food and water intake

Depending on the type of exposure, greater body weight was not always related to hyperphagia. We observed that the noise exposure alone induced hyperphagia; this is consistent with the literature (such as exposure of rodents to 85 dB for 9 days, described by Mavanji et al. These effects are probably due to the intensity or chronicity of the exposure, given that acute exposure (for 12 h) in another study did not induce modifications in the food intake (Parrish and Teske, 2017).

Conversely, RF-EMF exposure alone did not induce a variation in the total quantity of food ingested – only the pattern of intake was modified. In a previous study in our laboratory, we did not observe any significant variations in the total quantity of food ingested with RF-EMF exposure in the same condition (Pelletier et al., 2013). Thus, a chronic exposure to RF-EMF alone did not modify the food intake. Interestingly, in this previous study, the total food intake was greater when RF-EMF exposed animals were housed at an ambient temperature of 31 °C during one day (relative to non-exposed animals housed at the same temperature). However, in our present study, the total food intake in the group chronically exposed to RF-EMF and noise was similar to that in the control group - even though the food intake pattern was

different. The intake in the RF-EMF+noise group showed some characteristics of noise effects: i.e. a lower number of meals and a longer meal duration. In contrast to our starting hypothesis, exposure to both an RF-EMF and noise was not associated with accentuation of the effects of noise alone or of RF-EMF alone with regard to food intake variables.

Exposure to noise alone or to an RF-EMF alone induced a greater water intake than in controls. Alterations in water intake have not previously been studied in RF-EMF-exposed animals. Surprisingly (and as for the total food intake), the water intake of animals exposed to both an RF-EMF and noise was similar to that of controls and was lower than that of animals exposed to noise alone or an RF-EMF alone. To the best of our knowledge, this antagonistic effect of RF-EMF vs. noise has not previously been described for food intake or weight gain parameters.

#### *4.3 Sleep parameters*

Surprisingly, noise exposure (with and without RF-EMF exposure) did not induce any significant sleep alterations during the daytime-resting period (i.e. when noise was present) – suggesting that the sleep need was predominant and/or the environmental constraint was not great enough to reduce sleep duration. However, during the nighttime (active, non-noisy) period, the animals in the noise-only group displayed a lower mean duration of NREM sleep (with no change in TST), a shorter QW duration, a lower NREM/REM ratio and a higher AW and REM duration. Noise exposure alone induce a higher mean duration of NREM episodes, independently of the light and dark periods. The absence of a difference in TST is surprising, given that two other studies of noise-exposed rats observed a lower TST during the daytime-resting period (i.e. when noise was present) (Mavanji et al., 2013; Parrish and Teske, 2017). Even though these studies used much the same protocol as we did (i.e. noise during the light period, and sleep assessment during the dark and light periods), the overall period of exposure (85 dB, for 9 h during the daytime) was shorter (1 to 9 days).

During the second recording period (a one-hour period outside the noise exposure chamber), the noise-exposed group displayed a higher TST (8 min more than in a control group) and lower amount of AW (-10.2 min less than control group). This is a typical sleep rebound phenomenon, and indicates greater sleep pressure in this group even though it was not statistically significant during the light period with noise exposure. The absence of sleep perturbation might have been due to a strong circadian influence independently of sleep (Borbely 1982). It might also have been due to the time distribution of the sleep during the dark (resting) period; a greater TST was observed during the first part of the night - as soon as

noise exposure stopped – but did not achieve statistical significance over the whole 12 h dark period. The lack of significant active time after-effects on TST and the greater TST during the one hour-recording without noise exposure might be due to the difference in the time interval between sleep onset and the end of the noise exposure (30 min for the one-hour session but 60 min for the 24 h recording). Averaging TST over the 12 h active period might mask effects occurring in the first few hours of the following active period (Bach et al. 1991). The nighttime effects on the chosen sleep parameters also evidenced (in line with a rebound effect) the invisible effects of noise exposure during the daytime.

Regardless of the noise exposure, the animals exposed to RF-EMF with or without noise, showed a slight fragmentation of AW (a higher episode frequency and a lower mean duration of episodes) but no change in the total AW duration. However, the animals exposed to RF-EMF alone has a higher mean duration for NREM sleep episodes. Some literature data have highlighted changes in sleep architecture with RF-EMF exposure, although REM was mainly affected. Acute RF-EMF exposure (900 MHz for 8 h) in humans was associated with a shorter sleep latency and a significant lower duration of REM sleep (Mann and Röschke, 1996; Wagner et al., 1998). The chronic exposure performed in our laboratory (5 weeks) was associated with REM sleep fragmentation (Pelletier et al., 2013). However, the effects of RF-EMF on sleep remain subject to debate.

Co-exposure to both environmental constraints (RF-EMF and noise) resulted in values similar to those observed in controls for the 24 h-sleep recording and the 1 h (noise- and RF-EMF-free) sleep period. There are several possible explanation for this finding. Firstly, one can hypothesize that RF-EMFs protect against the harmful effects of noise exposure on sleep. Secondly, the addition of a second factor might increase the overall level of stress and thus trigger defense mechanisms that reduce sleep and homeostatic perturbations. A spectral analysis might facilitate the analysis of these putative effects and could provide additional insights.

## 5. Conclusion

Noise and RF-EMF are both environmental constraints but acted in different ways on the parameters studied here. Firstly, noise exposure was associated with a greater body weight gain and a higher food intake. Sleep parameters were altered indirectly (i.e. during the nighttime noise-free period), with a lower NREM/REM balance but no change in the TST. However, a rebound effect was observed (a higher TST, more REM sleep, and less AW) when animals slept during a 1 h noise- and EMF-free period - indicating that sleep pressure may have been increased.

The RF-EMF-only group showed a greater body weight gain. An assessment of sleep parameters during the 24 h recording highlighted the fragmentation of AW (even though no such effect was observed during the 1 h constraint-free period); this fragmentation argued in favor of a direct effect of the RF-EMF. Animals exposed to an RF-EMF appeared to reduce their energy expenditure by limiting REM sleep and increasing food intake.

Animals simultaneously to both noise and an RF-EMF were also heavier than controls, although there was no difference in the total food quantity. Hence, co-exposure to noise did not exacerbate the effects of RF-EMF exposure on juvenile rats. Indeed, most of the effects on sleep in animals exposed to noise alone or RF-EMF exposure alone disappeared in co-exposed animals. In contrast to our hypothesis, co-exposure did not accentuate the effects observed with single exposures; this contrasts with the results of other studies of co-exposures (i.e. ambient temperature changes and neuroinflammatory reactions) (Occelli et al., 2018; Pelletier et al., 2013).

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441



| Food and water intake parameters   | Control group | Noise-exposed group   | RF-EMF exposed group | RF-EMF+noise group   |
|------------------------------------|---------------|-----------------------|----------------------|----------------------|
| Total quantity of food (g)         | 18.5±2.4      | 21.2±3.0<br>***##     | 18.1±6.9 **          | 19.6±4.1***          |
| Total quantity of food/body weight | 0.066±0.006   | 0.073±0.008<br>***### | 0.066±0.007          | 0.068±0.007<br>trend |
| Number of meals                    | 12.2±2.2      | 13.4±2.6<br>**###     | 10.3±4.1<br>**       | 10.6±2.8<br>***      |
| Mean meal duration (s)             | 667.3±228.7   | 819.9±324.9<br>**     | 545.9±285.1<br>*###  | 806.9±375.2<br>**    |
| Total quantity of water (mL)/24 h  | 28.9±6.7      | 36.2±2.7<br>***#      | 33.4±9.0<br>***#     | 30.5±7.6             |

**Table 1:** Food and water intakes (mean ± SD) in control (n=12), noise-exposed (n=12), RF-EMF-exposed (n=11) and RF-EMF+noise-exposed (n=12) animals per 24 h. The total quantity of food eaten was expressed in grams and relative to the body weight. A Mann-Whitney test was used to compare each group with the control group (\* p<0.05 \*\* p<0.01, \*\*\* p<0.001, trend 0.1<p<0.05) and the RF-EMF+noise-exposed group (# p<0.05 ## p<0.01, ### p<0.001).

| Parameters           | Stages of vigilance | Period               | Control group  | Noise group    | RF-EMF group   | RF-EMF+noise group | Statistical analysis <sup>a</sup> |
|----------------------|---------------------|----------------------|----------------|----------------|----------------|--------------------|-----------------------------------|
| Total duration (min) | TST                 | Light (rest) period  | 469.3±3<br>6.0 | 485.8±3<br>6.2 | 491.5±3<br>5.6 | 481.9±42.7         |                                   |
|                      |                     | Dark (active) period | 284.8±2<br>9.9 | 267.4±3<br>6.4 | 288.2±3<br>0.7 | 272.1±53.5         |                                   |
|                      | NREM                | Light (rest) period  | 417.4±2<br>9.3 | 438.5±3<br>3.5 | 441.4±2<br>9.6 | 438.6±35.0         |                                   |
|                      |                     | Dark (active) period | 225.3±2<br>6.0 | 202.7±2<br>7.5 | 227.1±2<br>1.3 | 204.2±36.9         | Noise effect                      |

|                                    |      |                                |                |                     |                |            |                                         |
|------------------------------------|------|--------------------------------|----------------|---------------------|----------------|------------|-----------------------------------------|
|                                    |      | )<br>period                    |                |                     |                |            | p=0.01 <sup>b</sup>                     |
|                                    | REM  | Light<br>(rest)<br>period      | 51.9±14<br>.6  | 47.3±15<br>.1       | 50.2±17<br>.2  | 43.3±13.5  |                                         |
|                                    |      | Dark<br>(active<br>)<br>period | 59.5±8.<br>3   | 64.7±14<br>.0 trend | 61.1±15<br>.2  | 67.9±18.8  | Noise<br>effect<br>p=0.1 <sup>b</sup>   |
|                                    | AW   | Light<br>(rest)<br>period      | 152.3±2<br>5.4 | 138.8±2<br>8.3      | 149.1±3<br>0.1 | 139.2±37.1 |                                         |
|                                    |      | Dark<br>(active<br>)<br>period | 394.9±3<br>2.5 | 419.1±4<br>7.0      | 393.9±2<br>9.0 | 415.7±61.1 | Noise<br>effect<br>p=0.085 <sup>b</sup> |
|                                    | QW   | Light<br>(rest)<br>period      | 98.4±29<br>.2  | 95.4±20<br>.1       | 79.4±17<br>.8  | 98.8±18.4  |                                         |
|                                    |      | Dark<br>(active<br>)<br>period | 40.3±10<br>.3  | 33.5±11<br>.7       | 37.9±6.<br>7   | 32.2±9.2   | Noise<br>effect<br>p=0.029 <sup>b</sup> |
| Relative<br>duration (%<br>of TST) | NREM | Light<br>(rest)<br>period      | 89.0±2.<br>7   | 90.3±2.<br>9        | 89.9±3.<br>0   | 91.1±2.4   |                                         |
|                                    |      | Dark<br>(active<br>)<br>period | 79.0±2.<br>3   | 75.9±3.<br>7        | 79.0±3.<br>9   | 75.6±4.3   | Noise<br>effect<br>p=0.004 <sup>b</sup> |
|                                    | REM  | Light<br>(rest)<br>period      | 11.0±2.<br>7   | 9.7±2.9             | 10.1±3.<br>0   | 8.9±2.5    |                                         |
|                                    |      | Dark<br>(active<br>)<br>period | 21.0±2.<br>3   | 24.1±3.<br>7        | 21.0±3.<br>9   | 24.4±4.3   | Noise<br>effect                         |

|                                      |      |                                |              |              |              |          |                                                           |
|--------------------------------------|------|--------------------------------|--------------|--------------|--------------|----------|-----------------------------------------------------------|
|                                      |      | )<br>period                    |              |              |              |          | p=0.004 <sup>b</sup>                                      |
| Frequency<br>(ep)                    | NREM | Light<br>(rest)<br>period      | 22.3±3.<br>2 | 19.9±2.<br>8 | 19.4±3.<br>6 | 22.0±5.1 |                                                           |
|                                      |      | Dark<br>(active<br>)<br>period | 11.2±1.<br>7 | 8.7±2.2      | 11.2±1.<br>2 | 9.7±2.2  |                                                           |
|                                      | REM  | Light<br>(rest)<br>period      | 4.6±1.3      | 4.2±1.5      | 4.3±1.1      | 4.1±1.1  |                                                           |
|                                      |      | Dark<br>(active<br>)<br>period | 3.9±0.7      | 3.8±0.8      | 4.0±0.9      | 4.4±1.3  |                                                           |
|                                      | AW   | Light<br>(rest)<br>period      | 1.3±0.5      | 1.0±0.4      | 1.3±0.5      | 1.6±0.9  | RF-EMF<br>effect<br>p=0.029 <sup>c</sup>                  |
|                                      |      | Dark<br>(active<br>)<br>period | 1.5±0.4      | 1.6±0.9      | 2.0±0.8      | 1.7±0.5  |                                                           |
|                                      | QW   | Light<br>(rest)<br>period      | 21.0±3.<br>3 | 18.9±2.<br>6 | 18.1±3.<br>6 | 20.5±4.7 |                                                           |
|                                      |      | Dark<br>(active<br>)<br>period | 9.7±1.6      | 7.3±2.1      | 9.3±1.0      | 8.0±2.3  |                                                           |
| Mean<br>episode<br>duration<br>(min) | NREM | Light<br>(rest)<br>period      | 1.6±0.3      | 1.8±0.3      | 2.0±0.4      | 1.7±0.4  | RF-EMF<br>and noise<br>interaction<br>p=0.01 <sup>d</sup> |
|                                      |      | Dark<br>(active)               | 1.7±0.2      | 2.0±0.5      | 1.7±0.3      | 1.8±0.3  |                                                           |

|  |     |                                |              |               |              |          |                                          |
|--|-----|--------------------------------|--------------|---------------|--------------|----------|------------------------------------------|
|  |     | )<br>period                    |              |               |              |          |                                          |
|  | REM | Light<br>(rest)<br>period      | 1.0±0.2      | 1.0±0.3       | 1.0±0.3      | 0.9±0.2  |                                          |
|  |     | Dark<br>(active<br>)<br>period | 1.3±0.3      | 1.4±0.2       | 1.3±0.2      | 1.3±0.2  |                                          |
|  | AW  | Light<br>(rest)<br>period      | 10.9±3.<br>7 | 12.1±3.<br>6  | 10.8±3.<br>5 | 9.1±4.6  | RF-EMF<br>effect<br>p=0.012 <sup>c</sup> |
|  |     | Dark<br>(active<br>)<br>period | 23.9±7.<br>9 | 27.8±13<br>.2 | 18.0±6.<br>0 | 22.2±6.8 |                                          |
|  | QW  | Light<br>(rest)<br>period      | 0.4±0.0<br>8 | 0.4±0.0<br>7  | 0.4±0.0<br>4 | 0.4±0.07 |                                          |
|  |     | Dark<br>(active<br>)<br>period | 0.3±0.0<br>7 | 0.4±0.0<br>6  | 0.3±0.0<br>6 | 0.3±0.06 |                                          |

448

449 **Table 2:** Sleep parameters (mean ± SD) in control (n=12), noise-exposed (n=10), RF-EMF-exposed  
450 (n=11) and RF-EMF+noise-exposed (n=12) animals per 24 h, and split into light (rest/noise exposure)  
451 and dark (activity/no noise) periods. AW: active wakefulness; QW: quiet wakefulness; NREM: non-  
452 rapid-eye-movement; REM: rapid-eye-movement; TST: total sleep time. Statistical results from the  
453 analysis of variance.

454 <sup>a</sup>: Column indicate results from ANOVA analysis.

455 <sup>b</sup>: *p*-values represent significant effect of noise exposure, only during the dark period, whatever the  
456 presence of RF-EMF exposure.

457 <sup>c</sup>: *p*-values represent significant effect of RF-EMF exposure, whatever the considered period and the  
458 presence of noise exposure.

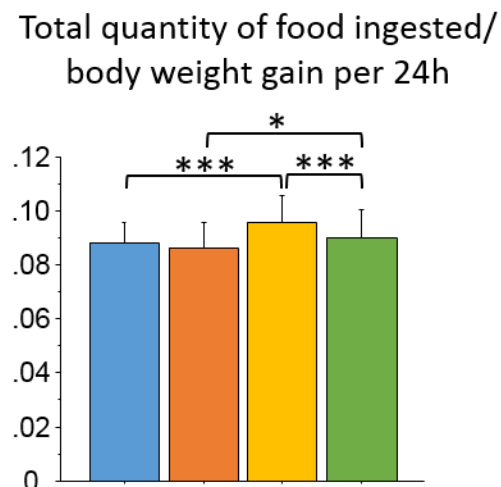
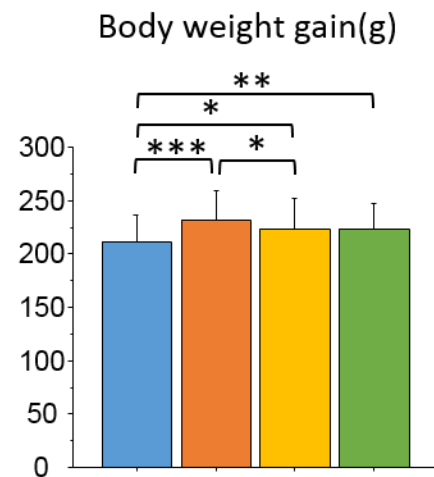
<sup>d</sup>: *p*-value represents significant interaction between the two factors: RF-EMF exposure and noise exposure.

|                              |       |      | Control group | Noise group | RF-EMF group | RF-EMF+noise group | Effect of noise <sup>a</sup> | Effect of RF-EMF <sup>b</sup> |
|------------------------------|-------|------|---------------|-------------|--------------|--------------------|------------------------------|-------------------------------|
| Total duration (min)         | sleep | TST  | 32.5±13.9     | 43.2±7.1    | 33.8±15.7    | 39.2±11.5          | p=0.03                       | NS                            |
|                              |       | NREM | 31.5±13.1     | 40.4±7.0    | 32.6±14.8    | 37.7±10.9          | p=0.05                       | NS                            |
|                              |       | REM  | 1.0±1.5       | 2.8±2.3     | 1.2±2.1      | 1.7±1.3            | p=0.05                       | NS                            |
|                              | wake  | AW   | 22.2±16.3     | 9.8±8.5     | 22.5±17.1    | 14.1±13.8          | p=0.02                       | NS                            |
|                              |       | QW   | 5.3±3.3       | 7.0±3.3     | 3.7±1.6      | 6.6±3.1            | p=0.01                       | NS                            |
| Relative duration (% of TST) | sleep | NREM | 97.4±3.5      | 93.6±5.6    | 97.2±4.3     | 96.3±2.8           | NS                           | NS                            |
|                              |       | REM  | 2.6±3.5       | 6.4±5.6     | 2.8±4.3      | 3.7±2.8            | NS                           | NS                            |
| Frequency (ep/h)             | sleep | NREM | 15.4±7.4      | 16.0±6.9    | 13.7±4.3     | 17.7±6.9           | NS                           | NS                            |
|                              |       | REM  | 1.2±1.7       | 3.8±2.6     | 2.0±2.4      | 3.2±2.1            | p=0.007                      | NS                            |
|                              | wake  | AW   | 2.4±1.6       | 1.4±0.8     | 2.8±2.3      | 1.5±1.2            | p=0.01                       | NS                            |
|                              |       | QW   | 13.1±6.9      | 14.9±6.8    | 10.8±4.5     | 16.1±6.5           | NS                           | NS                            |
| Mean episode duration (min)  | sleep | NREM | 2.0±0.9       | 3.0±1.3     | 2.5±1.3      | 2.4±1.1            | NS                           | NS                            |
|                              |       | REM  | 0.7±0.6       | 0.7±0.4     | 0.5±0.4      | 0.7±0.7            | NS                           | NS                            |
|                              | wake  | AW   | 9.4±12.6      | 5.7±6.1     | 13.3±17.0    | 9.0±6.6            | NS                           | NS                            |
|                              |       | QW   | 0.4±0.2       | 0.5±0.2     | 0.3±0.1      | 0.4±0.1            | NS                           | NS                            |

**Table 3:** Sleep parameters (mean ± SD) in control (n=12), noise-exposed (n=11), RF-EMF-exposed (n=10) and RF-EMF+noise-exposed (n=12) animals measured during a one-hour, RF-EMF- and noise-free. AW: active wakefulness; QW: quiet wakefulness; NREM: non-rapid-eye-movement; REM: rapid-eye-movement; TST: total sleep time. Statistical results from the analysis of variance.

<sup>a</sup>: *p*-values represent significant differences between the noise non-exposed group and the noise exposed group, whatever the presence of RF-EMF exposure (NS: non-significant).

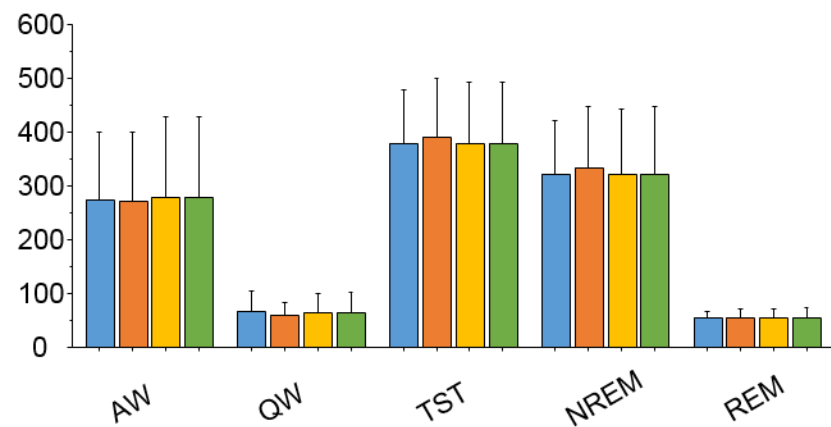
<sup>b</sup>: *p*-values represent significant differences between the RF-EMF non-exposed group and the RF-EMF exposed group, whatever the presence of noise exposure (NS: non-significant).



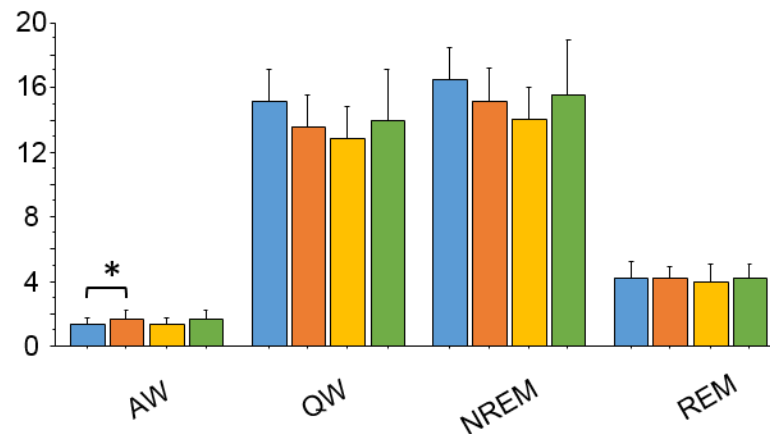
|               |                    |
|---------------|--------------------|
| Control group | RF-EMF group       |
| Noise group   | RF-EMF+noise group |

\* p<0,05 ; \*\* p<0,01 ; \*\*\* p<0,001

Total duration of sleep stages (min)



Frequency of episodes (épisode/h)



Mean duration of episodes (s)

